AIGR Function-Inference Stress Test — PP_4594 (Q88E72) vs GO:0004123

Target: PP_4594 / UniProt Q88E72, *Pseudomonas putida* KT2440 (NCBITaxon:160488) **Seed hypothesis (TreeGrafter/PANTHER, IEA, GO_REF:0000118):** PP_4594 has cystathionine gamma-lyase activity (GO:0004123, EC 4.4.1.1).

Executive Judgment

Verdict: PARTIALLY SUPPORTED / sibling-activity ambiguity (failure mode 1 — granularity / family-vs-subfamily).

PP_4594 is unambiguously a catalytically competent, fold-type I PLP-dependent enzyme of the **trans-sulfuration (Cys/Met metabolism) family, gamma subfamily**. The *catalytic machinery is intact* and the protein sits in the correct broad family, so the term is correct at the superfamily/catalytic-competence level. However, the **specific** assignment "cystathionine gamma-lyase" cannot be distinguished by sequence from its near-identical siblings **methionine gamma-lyase (EC 4.4.1.11)** and **cystathionine gamma-synthase (EC 2.5.1.48)**. There is no experimental assay of PP_4594.

- **Failure mode 2 (pseudo-enzyme): RULED OUT.** Catalytic PLP-Schiff-base Lys205 is conserved and correctly positioned; PLP cofactor annotated; fold intact.
- **Failure mode 3 (mis-placement into a distinct neighbor): RULED OUT.** It is clearly a *gamma*-subfamily enzyme, not a cystathionine beta-lyase (only 29% identity to MetC vs 39–44% to gamma members).
- **Failure mode 1 (granularity / sibling activity): APPLIES.** The propagated CGL term is one of several indistinguishable sibling activities.

Single most decisive piece of evidence: In a BLOSUM62 global alignment, PP_4594 is **equidistant (~40%) to all gamma-family activities**, with the single top characterized homolog being *methionine gamma-lyase* (44.3%), essentially tied with human CGL (43.9%) and above cystathionine gamma-synthase (38.8%) — i.e., sequence provides **no discrimination** of the

exact EC, while the family literature states CGS and CGL "have highly similar amino acid sequences"
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Independent Family / Function Assignment

- **Most likely specific function (from first principles):** a **pyridoxal-5′-phosphate-dependent, gamma-family carbon–sulfur lyase/synthase of the trans-sulfuration family** (PF01053 / IPR000277). The physiological substrate is not resolvable computationally.
- **Nearest characterized homolog:** *Pseudomonas putida* L-methionine gamma-lyase (P13254, 44.3% id), essentially tied with human cystathionine gamma-lyase (P32929, 43.9%).
- **GO term I would assign with confidence:** parent term **carbon–sulfur lyase activity (GO:0016846)** or the well-supported **pyridoxal phosphate binding (GO:0030170)** + **transsulfuration (GO:0019346)**. The seed term **GO:0004123** is of *correct granularity depth but names one specific sibling among several equally plausible ones* (same branch, specific; relationship to seed = **plausible but unverifiable sibling**).

Evidence Matrix

#	Citation / Source	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
1	UniProt Q88E72 (features/xrefs)	Database / computational	Qualifies	Is this a PLP trans-sulfuration enzyme?	PF01053 Cys_Met_Meta_PP; SSF53383 PLP transferase; PLP cofactor; catalytic Lys205	Reviewed annotation (IEA/ARBA)
2	This study — NW alignment, catalytic Lys	Computational / structural-evolutionary	Supports (not pseudo-enzyme)	Is catalysis retained?	Lys205 conserved & aligned to catalytic Lys of CGL/MGL/CGS/ CBL; conserved S-x-S-K motif	BLOSUM62 global align
3	This study — %identity table	Computational / evolutionary	Qualifies / competing	Which sibling activity?	~40% equidistant to MGL(44.3)/ CGL(43.9)/ OSHS(41.5)/ CGS(38.8); CBL only 29.4%	Characterize refs
4	PANTHER PTHR11808:SF15	Database (phylo)	Supports seed	Subfamily call	SF15 = cystathionine gamma-lyase	TreeGrafter source
5	KEGG K01758 (PP_4594)	Database (ortholog)	Supports seed	KO call	K01758 = cystathionine gamma-lyase EC 4.4.1.1	Independent of PANTHER
6	GenBank/EMBL AAN70167.1 name	Database	Competing	Product name	"cystathionine gamma-synthase"	Original author annotation
7	KEGG ppu gene set	Database (genomic context)	Qualifies (supports	Is PP_4594 the biosynthetic MetB?	No — dedicated metB=PP_0659 (K01739),	*P. putida* KT2440

#	Citation / Source	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
			catabolic CGL role)		metC=PP_4348 (K01760), metY=PP_2528	
8	P 41562399 (2026)	Review / structural	Qualifies	CGS vs CGL separability	"CGS and CGL have highly similar amino acid sequences"; specificity set by few pocket residues (E55, V232)	*L. plantarum*
9	P 28106980 (2017)	Structural / mutational	Qualifies	CGL↔MGL interconversion	Human CGL scaffold engineered into an L-Met-degrading enzyme (MGL activity)	Human CGL

Active-Site / Placement Analysis (provenance)

Catalytic PLP-Schiff-base lysine (PP_4594 Lys205) — conserved across all references:

Reference (characterized)	PP_4594 K205 aligns to	Activity
human CGL (P32929)	**K** (K212)	cystathionine gamma-lyase
yeast CYS3 (P31373)	**K**	cystathionine gamma-lyase
P. putida MGL (P13254)	**K**	methionine gamma-lyase
E. coli MetB (P00935)	**K**	cystathionine gamma-synthase
E. coli MetC (P06721)	**K**	cystathionine beta-lyase
OSHS metZ (P55218)	**K**	O-succinylhomoserine sulfhydrylase

Conserved PLP motif window (PP_4594 `DLVMHSASKYIGGHSDL` vs human CGL `DISMYSATKYMNGHSDV`) — the catalytic lysine and flanking PLP-binding scaffold are intact. **Catalytic residues are present and correctly spaced → the enzyme is catalytically competent (not a pseudo-enzyme).**

Subfamily placement (global %identity, BLOSUM62 NW):

```
MGL  P. putida   P13254  44.3%   (gamma: methionine gamma-lyase)  <-- top hit
CGL  human       P32929  43.9%   (gamma: cystathionine gamma-lyase = seed)
OSHS metZ        P55218  41.5%   (gamma: O-succinylhomoserine sulfhydrylase)
CGL  yeast CYS3  P31373  41.2%   (gamma: cystathionine gamma-lyase)
CGS  metB E.coli P00935  38.8%   (gamma: cystathionine gamma-synthase)
CBL  metC E.coli P06721  29.4%   (beta:  cystathionine beta-lyase)   <-- clearly excluded
```

Interpretation: gamma subfamily confirmed; exact sibling activity **unresolved** (all gamma activities within a 39–44% band, no outlier winner).

Iteration 2 — UPGMA phylogenetic placement (10 taxa, BLOSUM62 NW distances):

```
%identity of PP_4594 to reviewed references (sorted):
 CGS M. tuberculosis (P9WGB7)  45.7%   <-- single best hit = a gamma-SYNTHASE
 CGL rat           (P18757)    44.5%
 MGL P. putida     (P13254)    44.3%
 CGL human         (P32929)    43.9%
 CGL yeast CYS3    (P31373)    41.2%
 CGS E. coli metB  (P00935)    38.8%
 MGL Brevibacterium(A0A2H1K3G9)38.7%
 OAHS S. pombe     (O13326)    37.1%
 CBL E. coli metC  (P06721)    29.4%   <-- clear outgroup (beta-lyase)

UPGMA: PP_4594 first merges with MGL_Pputida (d=55.7), then with the
       CGL(mammal/yeast)+CGS(Mtb) clade (d=56.1); CBL is the outgroup.
```

The query sits **at the junction of the gamma clades with no clean assignment**: its single best sequence match is a cystathionine gamma-*synthase* (Mtb) and its nearest tree neighbour is a methionine gamma-*lyase* — neither is the seed's cystathionine gamma-*lyase*. This independently corroborates the sibling-activity ambiguity and again excludes the beta-lyase branch. (Caveat: UPGMA is an unrooted distance phenogram, not a bootstrapped ML tree.)

GO Curation Implications

Lead for curator verification: Treat GO:0004123 as a plausible-but-low-confidence computational sibling call; do NOT assert it as directly demonstrated. Recommended actions, in order of preference:

1. **Generalize / safe retain:** keep the better-supported parent/companion terms — **pyridoxal phosphate binding (GO:0030170)**, **transsulfuration (GO:0019346)**, and, for MF, consider the parent **carbon–sulfur lyase activity (GO:0016846)** — all of which are robustly supported by fold and catalytic-residue evidence.
2. **If a specific MF term is required:** retain GO:0004123 **flagged IEA/low-confidence**, with an explicit note that methionine gamma-lyase (GO:0018826 / EC 4.4.1.11) and cystathionine gamma-synthase (GO:0003962 / EC 2.5.1.48) are indistinguishable siblings and that a separate *bona fide* metB (PP_0659) exists.
3. **Do not** upgrade the IEA to a more specific or experimental-equivalent evidence code without assay.

(Per instructions, "protein binding" is explicitly avoided as a recommendation.)

Conflicts, Knowledge Gaps, Discriminating Tests

- **Database conflict:** PANTHER SF15 + KEGG K01758 say *cystathionine gamma-lyase*; GenBank product name says *cystathionine gamma-synthase*; the single closest characterized sequence is *methionine gamma-lyase*. These three sibling activities cannot be separated from sequence.
- **Paralog/context:** *P. putida* KT2440 encodes dedicated metB (PP_0659) and metC (PP_4348) for forward (biosynthetic) transsulfuration, so PP_4594 is most plausibly a **catabolic/spare gamma-elimination enzyme** — consistent with (but not proof of) a CGL/MGL role rather than biosynthetic CGS.
- **Key uncertainty for curation:** no experimental characterization of PP_4594 exists; all evidence is IEA/computational, and two of the supporting calls (PANTHER, KEGG) are themselves automated, so they are not independent confirmation of the TreeGrafter graft.
- **Most efficient discriminating test:** a simple in-vitro PLP enzyme assay on recombinant PP_4594 measuring (a) cystathionine → cysteine + α-ketobutyrate + NH3 (CGL), (b) L-methionine → methanethiol + α-ketobutyrate + NH3 (MGL), and (c) O-acyl-homoserine + cysteine → cystathionine (CGS). Alternatively, an AlphaFold-model active-site pocket comparison of the substrate-specificity residues (e.g., the positions equivalent to *L. plantarum* CGS E55/V232 per

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) would computationally narrow the call.

Supported / Refuted Hypotheses

- **Supported:** PP_4594 is a catalytically intact PLP trans-sulfuration enzyme of the gamma subfamily (not a pseudo-enzyme; not a beta-lyase).
- **Partially supported:** the seed GO:0004123 (CGL) — correct family and plausible, but one of several indistinguishable siblings; not directly demonstrated.
- **Refuted:** that the specific CGL EC can be confirmed from sequence alone (it cannot); that PP_4594 is the organism's biosynthetic cystathionine gamma-synthase (that is PP_0659).

Limitations

Global NW identity with BLOSUM62 and a UPGMA distance phenogram (not a profile/HMM, bootstrapped ML tree, or structural pocket analysis); no biochemical assay accessible; AlphaFold pocket-residue comparison and a formal MSA of specificity residues were not run and would further refine the sibling call. The ~40–46% identities to all gamma siblings lie in the twilight zone for exact substrate/EC transfer, which is the fundamental reason the specific activity cannot be pinned computationally.